

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Java Spring Boot, Neo4j, and PostgreSQL stack.
- Authored an AWS deployment script and managed the environment as code with Terraform, replacing a manual 30-minute multi-service bring-up with a single-command, version-controlled provision.

Secure and Efficient Autonomous Systems Lab

Undergraduate Research Assistant

May 2026 – Present

East Lansing, MI

- Ran the node graph byte-identical in Gazebo and on a physical RoboMaster EP Core, driving the real robot over a custom C++ WiFi driver at a 30 Hz control loop with sub-20ms command latency.
- Built the verifier as a C++ ROS2 node running per-step geometric and semantic checks on each motion plan, integrating nav2's MPPI controller rather than hand-rolling one to keep the C++ work on security, not motion control.

Claude Builder Club @ MSU

Co-Founder & Vice President

Oct 2025 – Present

East Lansing, MI

- Co-founded and grew an MSU engineering community from zero to 150+ members, replacing a sprawl of spreadsheets, Slack threads, and manual invites with a single club-operations platform I designed and built.
- Automated member onboarding so an accepted applicant is provisioned into the right GitHub teams, Slack channels, Discord roles, and Google Drive access in a single flow, cutting onboarding from 15 minutes to seconds.
- Tuned React Query cache tiers by data volatility across the dashboards, caching stable roles longer than fast-changing application queues to cut redundant refetches while keeping live data fresh.

PROJECTS

Dadei | *React, FastAPI, Python, Redis*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Built a React client that streams microphone audio up over a WebSocket with client-side Web Audio processing, receiving incremental live captions back in under 300ms as the backend transcribes each utterance.
- Trained a custom wake-word model to 90
- Decoupled background workers from the live FastAPI service behind a Redis Streams event bus with consumer groups, dead-letter routing after 5 retries, and stale-claim recovery, giving at-least-once delivery that pub/sub dropped.

fliks | *Go, PostgreSQL*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Built a self-hosted video transcoding pipeline with a concurrent Go worker that pulls upload jobs, transcodes, and writes processed video back to object storage, owning the media path end to end instead of renting a managed service.
- Coordinated transcode jobs through a Postgres-backed queue using SELECT FOR UPDATE SKIP LOCKED instead of bolting on Redis or SQS, keeping the whole system on one datastore and one self-provisioned EC2 box.

TECHNICAL SKILLS

Languages Python, Go, JavaScript, TypeScript, Java, C++, SQL
Frameworks React, FastAPI, Spring Boot
Databases PostgreSQL, Neo4j
Robotics ROS2, Gazebo